



Laboratoire 00 : Infrastructure – Docker, Version Control, CI/CD

Cours	LOG430 – Architecture Logiciel
Session	ÉTÉ 2026
N° de laboratoire	00
Groupe	02
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Question 1 : Si l'un des tests échoue à cause d'un bug, comment pytest signale-t-il l'erreur et aide-t-il à la localiser ? Rédigez un test qui provoque volontairement une erreur, puis montrez la sortie du terminal obtenue.

Quand un test échoue, pytest affiche le message en capture d'écran ci-dessous :

```
# pytest
===== test session starts =====
platform linux -- Python 3.11.15, pytest-9.0.3, pluggy-1.6.0
rootdir: /app
configfile: pyproject.toml
collected 2 items

src/tests/test_calculator.py .F

===== FAILURES =====
test_addition
def test_addition():
    calc = Calculator()
    assert calc.addition(2, 3) == 7
E       assert 5 == 7
E         + where 5 = addition(2, 3)
E         + where addition = <calculator.Calculator object at 0x77a94128d810>.addition
```

Question 2 : Que fait GitHub pendant les étapes de « setup » et « checkout » ? Veuillez inclure la sortie du terminal GitHub CI dans votre réponse.

Pendant checkout, GitHub :

- télécharge ton dépôt GitHub
- récupère ton code source
- le place dans l'environnement virtuel (runner)

```
✓ Checkout dépôt

1 ▶ Run actions/checkout@v3
14 Syncing repository: leben89/log430-labo0-E26
15 ▶ Getting Git version info
19 Temporarily overriding HOME='/home/runner/work/_temp/79fc2e6f-bca7-4b8c-b3d6-ed63d727d861' before making global git config changes
20 Adding repository directory to the temporary git global config as a safe directory
21 /usr/bin/git config --global --add safe.directory /home/runner/work/log430-labo0-E26/log430-labo0-E26
22 Deleting the contents of '/home/runner/work/log430-labo0-E26/log430-labo0-E26'
23 ▶ Initializing the repository
40 ▶ Disabling automatic garbage collection
42 ▶ Setting up auth
48 ▶ Fetching the repository
119 ▶ Determining the checkout info
120 ▶ Checking out the ref
124 /usr/bin/git log -1 --format='%H'
125 'e1d56fce6df3cc4efa929af02c75491502485506'
```

```
deploy
succeeded 3 days ago in 7s

> ✓ Set up job

✓ Checkout dépôt

1 ▶ Run actions/checkout@v3
14 Syncing repository: leben89/log430-labo0-E26
15 ▶ Getting Git version info
19 Temporarily overriding HOME='/home/runner/actions-runner/_work/_temp/1eece70b-c27a-438d-ae66-561a90ac43b2' before making global git config changes
20 Adding repository directory to the temporary git global config as a safe directory
21 /usr/bin/git config --global --add safe.directory /home/runner/actions-runner/_work/log430-labo0-E26/log430-labo0-E26
22 /usr/bin/git config --local --get remote.origin.url
23 https://github.com/leben89/log430-labo0-E26
24 ▶ Removing previously created refs, to avoid conflicts
31 /usr/bin/git submodule status
32 ▶ Cleaning the repository
36 ▶ Disabling automatic garbage collection
38 ▶ Setting up auth
44 ▶ Fetching the repository
47 ▶ Determining the checkout info
48 ▶ Checking out the ref
52 /usr/bin/git log -1 --format='%H'
53 'e1d56fce6df3cc4efa929af02c75491502485506'
```

Pendant setup, GitHub :

- installe la bonne version de Python
- configure l'environnement d'exécution
- prépare pip

build
succeeded 3 days ago in 12s

✓ Set up job

1

Current runner version: '2.334.0'

2

▶ Runner Image Provisioner

9

▶ Operating System

13

▶ Runner Image

18

▶ GITHUB_TOKEN Permissions

22

Secret source: Actions

23

Prepare workflow directory

24

Prepare all required actions

25

Getting action download info

26

Download action repository 'actions/checkout@v3' (SHA:f43a0e5ff2bd294095638e18286ca9a3d1956744)

27

Download action repository 'actions/setup-python@v4' (SHA:7f4fc3e22c37d6ff65e88745f38bd3157c663f7c)

28

Complete job name: build

deploy

succeeded 3 days ago in 7s

✓ Set up job

1

Current runner version: '2.334.0'

2

Runner name: 'vm-eric-log430'

3

Runner group name: 'Default'

4

Machine name: 'vm-eric-log430'

5

▶ GITHUB_TOKEN Permissions

9

Secret source: Actions

10

Prepare workflow directory

11

Prepare all required actions

12

Getting action download info

13

Download action repository 'actions/checkout@v3' (SHA:f43a0e5ff2bd294095638e18286ca9a3d1956744)

14

Complete job name: deploy

Question 3 : Quel type d'informations pouvez-vous obtenir via la commande top ? Veuillez donner quelques exemples. Veuillez inclure la sortie du terminal dans votre réponse.

La commande top permet de surveiller en temps réel l'utilisation des ressources système. Elle affiche notamment l'utilisation du processeur (CPU), de la mémoire (RAM), ainsi que la liste des processus actifs avec leur consommation respective.

Par exemple, elle permet d'identifier les processus qui consomment le plus de CPU ou de mémoire, de voir la charge moyenne du système, ainsi que le nombre de tâches actives ou en sommeil. Ainsi, top est un outil essentiel pour diagnostiquer les performances et surveiller l'état du système en production.

1. Utilisation CPU

- quels processus consomment le plus de CPU
- ex: python, docker, systemd

2. Utilisation mémoire (RAM)

- combien de mémoire chaque processus utilise
- mémoire totale utilisée / libre

3. Processus actifs

- liste des processus en cours
- PID (identifiant processus)
- utilisateur (root, ubuntu, etc.)

4. Temps CPU

- combien de temps chaque processus a utilisé le CPU

5. Charge système

```
PS D:\ETS\Session 9 été 2026\LOG430-02\Laboratoires\Labo_00\log430-labo0-E26> ssh -i ~/.ssh/1xd_key root@10.194.32.158
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Tue May 12 04:38:11 UTC 2026

System load:  0.0          Processes:            85
Usage of /:   22.3% of 17.87GB   Users logged in:    0
Memory usage: 10%          IPv4 address for enp5s0: 10.194.32.158
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

64 updates can be applied immediately.
52 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 11 22:37:28 2026 from 10.192.175.139
root@vm-eric-log430:~# top
top - 04:39:37 up 7:26, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 82 total, 1 running, 81 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.3 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.3 st
MiB Mem : 3587.0 total, 97.5 free, 209.9 used, 3279.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 3310.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2681	root	20	0	1794188	52968	34356	S	0.3	1.4	0:32.31	containerd
5929	root	20	0	10872	3908	3200	R	0.3	0.1	0:00.20	top
1	root	20	0	167432	13016	8460	S	0.0	0.4	0:07.11	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	slub_flushwq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq